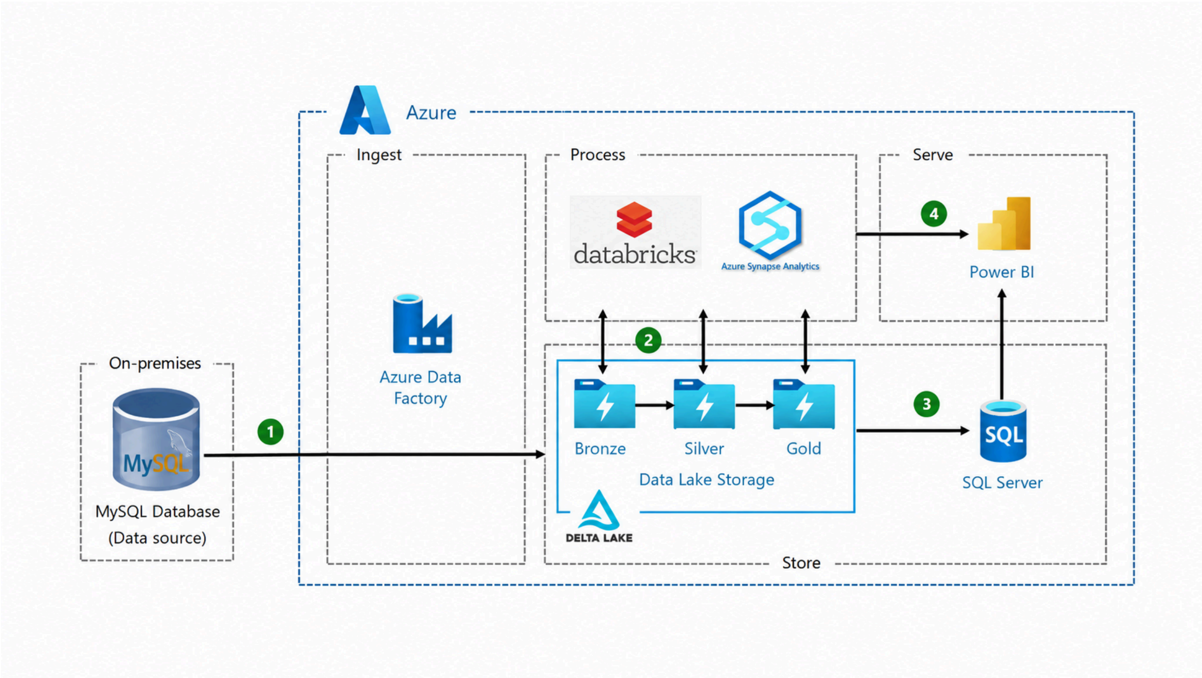
Azure Lakehouse Analytics Platform



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1. Introduction

The Azure Lakehouse Analytics Platform is a cloud-based end-to-end data engineering and analytics solution developed using Microsoft Azure services. The platform is designed to ingest, process, store, and visualize enterprise data efficiently using a modern Medallion Lakehouse Architecture.

The system integrates multiple Azure technologies including Azure Data Factory, Azure Data Lake Storage Gen2, Azure Databricks, Azure Synapse Analytics, Azure SQL Server, and Power BI to build a scalable and reliable analytics ecosystem.

The architecture supports structured data ingestion from MySQL databases and enables transformation of raw data into business-ready insights.

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2. Objective

The main objective of this project is to create a scalable and automated cloud-based data pipeline that:

- Extracts data from MySQL databases
- Processes and transforms data efficiently
- Stores data using Medallion Architecture
- Supports analytics and reporting
- Provides business insights using Power BI dashboards

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3. Architecture Overview

The project follows the Medallion Architecture pattern using three storage layers:

- Bronze Layer — Raw Data
- Silver Layer — Cleaned and Processed Data
- Gold Layer — Business Ready Data

The complete workflow includes:

1. Data ingestion from MySQL database
2. Data storage in Azure Data Lake
3. Data transformation using Databricks and Synapse Analytics
4. Data serving through SQL Server
5. Visualization using Power BI

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4. System Architecture Diagram

Architecture Flow

MySQL Database
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Azure Data Factory
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Azure Data Lake Storage (Bronze → Silver → Gold)
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Azure Databricks + Azure Synapse Analytics
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Azure SQL Server
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Power BI Dashboards

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5. Technologies Used

• MySQL — Source Database
• Azure Data Factory — Data Ingestion and Orchestration
• Azure Data Lake Storage Gen2 — Cloud Storage
• Delta Lake — Reliable Data Storage Layer
• Azure Databricks — Data Processing and Transformation
• Azure Synapse Analytics — Analytical Processing
• Azure SQL Server — Structured Data Serving
• Power BI — Reporting and Dashboard Visualization

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6. Detailed Workflow Explanation

Step 1 — Data Source (MySQL Database)

The pipeline starts with a MySQL database acting as the primary data source. The database contains operational or transactional business data.

Responsibilities

• Store source data
• Maintain transactional records
• Provide data for analytics processing

Advantages

• Structured relational storage
• Easy integration with Azure services
• Reliable data management

Step 2 — Data Ingestion using Azure Data Factory

Azure Data Factory (ADF) is used to extract data from the MySQL database and load it into Azure Data Lake Storage.

Responsibilities

• Data extraction
• Pipeline scheduling
• Workflow orchestration
• Data movement automation
• Monitoring and logging

Benefits

• Automated ETL/ELT workflows
• Scalable pipeline management
• Integration with cloud services

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7. Data Lake Storage and Medallion Architecture

The project uses Azure Data Lake Storage Gen2 with Delta Lake implementation.

The storage is divided into three layers:

7.1 Bronze Layer — Raw Data Layer

The Bronze layer stores raw data exactly as received from the MySQL source.

Characteristics

• Raw and unprocessed data
• Historical data preservation
• Immutable storage

Purpose

• Maintain source data integrity
• Support auditing and recovery

7.2 Silver Layer — Cleaned Data Layer

The Silver layer contains cleaned, validated, and transformed data.

Operations Performed

• Null handling
• Deduplication
• Data cleansing
• Data standardization
• Schema validation

Purpose

• Improve data quality
• Prepare datasets for analytics

7.3 Gold Layer — Business Layer

The Gold layer contains business-ready and analytics-optimized datasets.

Operations Performed

- KPI generation
- Aggregation
- Metric calculation
- Business transformations

Purpose

- Reporting
- Dashboard creation
- Business analytics

8. Data Processing Layer

Azure Databricks

Azure Databricks is used for distributed data processing using Apache Spark.

Responsibilities

- Large-scale data transformation
- ETL processing
- Distributed computing
- Batch analytics

Advantages

- High-speed processing
- Scalability
- Spark-based analytics

Azure Synapse Analytics

Azure Synapse Analytics provides enterprise-level analytical processing.

Responsibilities

- SQL analytics
- Data warehousing
- Big data querying
- Analytical reporting

Advantages

- High-performance querying
- Unified analytics environment
- Enterprise scalability

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9. Serving Layer — Azure SQL Server

Processed Gold-layer datasets are loaded into Azure SQL Server for efficient querying and reporting.

Responsibilities

- Structured data serving
- Query optimization
- Reporting support

Advantages

- Fast SQL queries
- Reliable relational storage
- Easy Power BI integration

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10. Visualization Layer — Power BI

Power BI is used to create interactive dashboards and reports from the processed datasets.

Features

- Interactive dashboards
- KPI visualization
- Real-time analytics
- Business intelligence reporting

Benefits

- Better decision making
- Data-driven insights
- User-friendly visualization

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11. End-to-End Data Flow

Phase 1 — Data Ingestion

Data is extracted from MySQL databases using Azure Data Factory.

Phase 2 — Data Storage

The ingested data is stored in Azure Data Lake Storage using Bronze, Silver, and Gold layers.

Phase 3 — Data Processing

Azure Databricks and Synapse Analytics process and transform the data.

Phase 4 — Data Serving

Curated business-ready data is loaded into Azure SQL Server.

Phase 5 — Data Visualization

Power BI dashboards are created for analytics and reporting.

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12. Key Features

- End-to-end cloud-native pipeline
- Automated ETL/ELT workflows
- Medallion Lakehouse Architecture
- Scalable distributed processing
- Centralized cloud storage
- Interactive BI dashboards
- Enterprise analytics support

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13. Advantages of the System

• Better data quality and governance
• Automated workflow orchestration
• High-performance distributed processing
• Reliable and scalable cloud storage
• Enterprise-level analytics support
• Optimized reporting performance
• Interactive business visualization

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14. Use Cases

• Enterprise Analytics
• Data Warehousing
• Business Intelligence Systems
• Real-Time Reporting
• Customer Analytics
• Financial Analytics
• Operational Dashboards

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15. Conclusion

The Azure Lakehouse Analytics Platform provides a scalable, reliable, and efficient solution for enterprise data engineering and analytics. By integrating Azure cloud services with Medallion Architecture, the platform enables organizations to transform raw operational data into meaningful business insights through automated ingestion, processing, storage, and visualization workflows.

The project demonstrates modern cloud data engineering practices and showcases the implementation of a complete analytics ecosystem using Microsoft Azure technologies.